SEQUENCE 4

# Real numbers and intervals

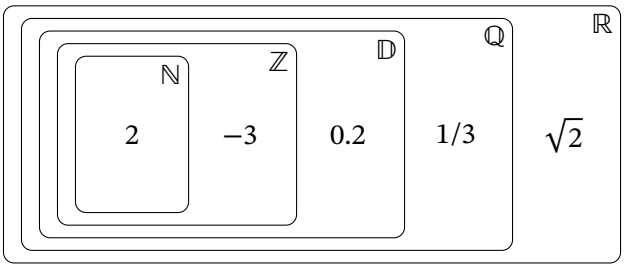
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## 4.1 Sets of numbers

**Key point · Definitions.**

- $\mathbb{N}$: natural numbers; $\mathbb{Z}$: integers.
- $\mathbb{D}$: decimal numbers, of the form $a/10^n$ with $a \in \mathbb{Z}$, $n \in \mathbb{N}$.
- $\mathbb{Q}$: rational numbers, of the form $a/b$ with $a, b \in \mathbb{Z}$, $b \neq 0$.
- $\mathbb{R}$: real numbers, represented by the abscissae of the points of a graduated line. A real number that is not rational is **irrational**.

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{D} \subsetneq \mathbb{Q} \subsetneq \mathbb{R}.$$




**Example 1.** Give the smallest of the five sets containing each of the numbers: $-95$; $7.29$; $5/3$; $\sqrt{25}$; $\sqrt{2}$; $\pi$; $0.0002$.

**Solution.** $-95 \in \mathbb{Z}$; $7.29 \in \mathbb{D}$; $5/3 \in \mathbb{Q}$; $\sqrt{25} = 5 \in \mathbb{N}$; $\sqrt{2}, \pi \in \mathbb{R} \setminus \mathbb{Q}$; $0.0002 = 2/10^4 \in \mathbb{D}$. The irrationality of $\pi$ is accepted here.

## 4.2 Two proofs by contradiction

**Method · Proof by contradiction.** To prove a statement, assume its negation, deduce a contradiction, then conclude that the statement is true.

**Key point · Property.** $\frac{1}{3}$ is not a decimal number.

**Proof to write.**

**Solution.** Suppose $1/3 = a/10^n$ with $a \in \mathbb{Z}$, $n \in \mathbb{N}$. Then $10^n = 3a$ would be a multiple of 3. Yet $10 \equiv 1$ modulo 3, so $10^n$ leaves remainder 1 on division by 3 (the divisibility rule for 3 can also be used). Contradiction. Hence $1/3 \notin \mathbb{D}$.

**Key point · Property.** $\sqrt{2}$ is irrational.

**Proof to write.**

**Solution.** Suppose $\sqrt{2} = p/q$ with $p, q \in \mathbb{N}$, $q \neq 0$, and $p/q$ in lowest terms. Then $p^2 = 2q^2$, so $p^2$ is even. If $p$ were odd, its square would be odd: so $p$ is even. Write $p = 2k$. We get $4k^2 = 2q^2$, then $q^2 = 2k^2$; so $q$ is even in turn. Thus 2 divides $p$ and $q$, contradicting irreducibility. Hence $\sqrt{2} \notin \mathbb{Q}$.

**Teacher note.** Have students identify separately the starting hypothesis, the implication used by contraposition on parity, and the final contradiction. Do not present a calculator's decimal expansion as a proof of irrationality.

## 4.3 Intervals of the real line

**Key point · Notation.** For two real numbers $a < b$, intervals describe the real numbers lying between two endpoints, with or without the endpoints. A square bracket includes the endpoint, a parenthesis excludes it. The symbols $-\infty, +\infty$ are not real numbers and always take a parenthesis.

| Condition | Interval | Description |
|---|---|---|
| $a \le x \le b$ | $[a, b]$ | both endpoints included |
| $a < x \le b$ | $(a, b]$ | $a$ excluded, $b$ included |
| $a \le x < b$ | $[a, b)$ | $a$ included, $b$ excluded |
| $a < x < b$ | $(a, b)$ | both endpoints excluded |
| $x \ge a$ | $[a, +\infty)$ | greater than or equal to $a$ |
| $x > a$ | $(a, +\infty)$ | strictly greater than $a$ |
| $x \le b$ | $(-\infty, b]$ | less than or equal to $b$ |
| $x < b$ | $(-\infty, b)$ | strictly less than $b$ |

$$\mathbb{R} = (-\infty, +\infty), \quad \mathbb{R}^* = (-\infty, 0) \cup (0, +\infty).$$

**Watch out.** The strictly negative real numbers form $(-\infty, 0)$: 0 is excluded. $[a, a] = \{a\}$; $(a, a) = \varnothing$.

**Example 2.** Write as an interval then represent: $2 \le x \le 5$; $x > 4$; $-6 < x \le 2$; $x < 0$.

**Solution.** $[2, 5]$; $(4, +\infty)$; $(-6, 2]$; $(-\infty, 0)$. On the number line, mark the endpoints, then show each one as included or excluded.

## 4.4 Intersection and union of intervals

**Key point · Definition.** $I \cap J$ corresponds to the condition "belonging to $I$ **and** to $J$"; $I \cup J$ corresponds to "belonging to $I$ **or** to $J$". A union of intervals is not always an interval.

**Example 3.** With $I = [0, 12]$ and $J = [3, 20]$, determine $I \cap J$ and $I \cup J$.

**Solution.** $I \cap J = [3, 12]$ and $I \cup J = [0, 20]$.

**Example 4.** Complete, justifying with representations on a number line.

| $I$ | $J$ | $I \cap J$ | $I \cup J$ |
|---|---|---|---|
| $(-4, 3]$ | $[1, 5]$ | $[1, 3]$ | $(-4, 5]$ |
| $(-\infty, 2]$ | $[-4, +\infty)$ | $[-4, 2]$ | $\mathbb{R}$ |
| $(-\infty, 7]$ | $[7, +\infty)$ | $\{7\}$ | $\mathbb{R}$ |
| $(-\infty, 5]$ | $[-3, +\infty)$ | $[-3, 5]$ | $\mathbb{R}$ |
| $(-2, 0)$ | $(0, 3]$ | $\varnothing$ | $(-2, 0) \cup (0, 3]$ |

**Solution.** For each intersection, keep the points common to both representations; for each union, keep all the points of at least one. In the last row, 0 belongs to neither interval: the union cannot be replaced by $(-2, 3]$.

**Fluency · 5-10 min.** Simplify $\overrightarrow{AB} + \overrightarrow{BC}$; solve $3x - 6 = 0$; calculate $10^{-2}$.
**Answer.** $\overrightarrow{AC}$; $x = 2$; $0.01$.

## 4.5 Absolute value and distance

**Key point · Definition and properties.** $|x|$ is the distance from the real number $x$ to 0; $|x| \geq 0$ and

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

The distance between $a$ and $b$ is $|b-a|$. For $a \in \mathbb{R}$ and $r \geq 0$:

$$|x-a| \leq r \iff x \in [a-r, a+r], \qquad |x-a| < r \iff x \in (a-r, a+r).$$

**Example 5.** Calculate $|3.2|$, $|-1.6|$, $|\sqrt{3}-2|$, then the distances between 2 and 5, 2 and $-5$, $-2$ and $-5$.

**Solution.** $3.2$; $1.6$; $2-\sqrt{3}$ since $\sqrt{3} < 2$. The distances are 3, 7 and 3.

**Example 6.** Solve: $|x-7/3| = 14$; $|x-2| \leq 3$; $|x+3| < 5$.

**Solution.** For the first equation, $x-7/3 = 14$ or $x-7/3 = -14$, so $S = \{-35/3, 49/3\}$. Then $|x-2| \leq 3 \iff x \in [-1, 5]$. Finally $|x+3| = |x-(-3)| < 5 \iff x \in (-8, 2)$.

**Example 7.** Write $x \in [4, 10]$ as an inequality with an absolute value.

**Solution.** The centre is $(4+10)/2 = 7$ and the radius $(10-4)/2 = 3$: $|x-7| \leq 3$.

## 4.6 Bounds and approximations

**Key point · Vocabulary.** To bound $x$ is to give $a \le x \le b$; the width is $b - a$. An approximation from below is less than or equal to $x$; an approximation from above is greater than or equal to it. Rounding chooses the nearest value at the required precision. Significant figures are counted from the first nonzero digit; the displayed precision must suit the context.

**Example 8.** Bound $\sqrt{2}$ between two consecutive hundredths. Give its rounding to the nearest hundredth. Round 0.004567 to three significant figures, then 12345 to three significant figures in scientific notation.

**Solution.** $1.41^2 = 1.9881 < 2 < 2.0164 = 1.42^2$, so $1.41 < \sqrt{2} < 1.42$ (width 0.01). The rounding to the nearest hundredth is 1.41. The other roundings are 0.00457 and $1.23 \times 10^4$.

```
def encadre_racine2(p):
    # p entier naturel ; calculs entiers exacts
    q = 10**p
    k = 0
    while (k+1)**2 <= 2*q*q:
        k = k+1
    return k/q, (k+1)/q
```

**Example 9.** Explain why this function bounds $\sqrt{2}$. What does it return for $p = 2$? Why does it terminate?

**Solution.** The loop looks for the largest integer $k$ such that $k^2 \le 2q^2$. On exit, $k^2 \le 2q^2 < (k+1)^2$, so $k/q \le \sqrt{2} < (k+1)/q$. For $p = 2$, it returns $(1.41, 1.42)$. Since $(2q)^2 > 2q^2$, the counter eventually reaches a value that makes the test false. The final divisions are floating-point approximations; the square tests are exact.